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v. Tannins in *Mimosa pudica*.

vi. Raphide crystals in *Dieffenbachia*.

vii. Heavy metals in Mustard (*Brassica*).

viii. Latex in Rubber tree and *Dandelion stem*.

Cell Inclusions

The cell inclusions are the non-living materials present in the cytoplasm. They are organic and inorganic compounds.

Inclusions in prokaryotes

In prokaryotes, reserve materials such as phosphate granules, cyanophycean granules, glycogen granules, poly β-hydroxy butyrate granules, sulphur granules, carboxysomes and gas vacuoles are present. Inorganic inclusions in bacteria are polyphosphate granules (volutin granules) and sulphur granules. These granules are also known as **metachromatic granules.**

Inclusions in Eukaryotes

- Reserve food materials: Starch grains, glycogen granules, aleurone grains, fat droplets
- Secretions in plant cells are essential oil, resins, gums, latex and tannins
- **Inorganic crystals** – plant cell have calcium carbonate, calcium oxalate and silica
- **Cystolith** – hypodermal leaf cells of *Ficus bengalensis*, calcium carbonate
- **Sphaeraphides** – star shaped calcium oxalate, *Colocasia*
- **Raphides** – calcium oxalate, *Eichhornia*
- **Prismatic crystals** – calcium oxalate, dry scales of *Allium cepa*

6.7. Nucleus

Nucleus is an important unit of cell which control all activities of the cell. Nucleus holds the hereditary information. It is the largest among all cell organelles. It may be spherical, cuboidal, ellipsoidal or discoidal.

It is surrounded by a double membrane structure called **nuclear envelope**, which has the inner and outer membrane. The inner membrane is smooth without ribosomes and the outer membrane is rough by the presence of ribosomes and is continues with irregular and infrequent intervals with the endoplasmic reticulum. The membrane is perforated by pores known as **nuclear pores** which allows materials such as mRNA, ribosomal units, proteins and other macromolecules to pass in and out of the nucleus. The pores enclosed by circular structures called **annuli**. The pore and annuli forms the **pore complex**. The space between two membranes is called **perinuclear space**.

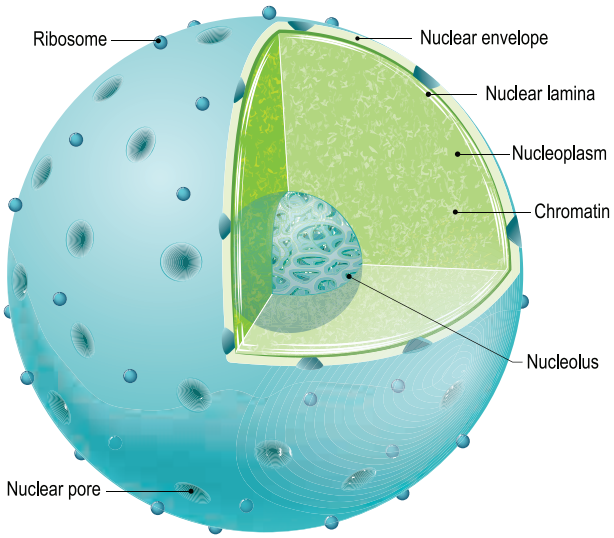


Figure 6.25: Structure of a Nucleus

Nuclear space is filled with **nucleoplasm**, a gelatinous matrix has uncondensed